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Chapter 17
Mixing in Bluffs: Finite Pot [0,1] Games

For the games in Chapter 16, we made a special rule that does not normally apply to poker situations - that no folding was allowed. In practice, we can sometimes encounter situations analogous to this when the pot is very large. One of the important features of these games is that there is no bluffing - the betting regions for both players are defined only in terms of the strongest hands and are designed only to extract value with those strongest hands.

However, bluffing is a key element of poker, and with its introduction to the game we obtain much more complexity, and additionally better approximations and insight into real poker. In this chapter, we will consider full-street games with finite pots; the primary modification that we make to these games compared to no-fold games is the introduction of bluffing regions that correspond to the value betting and raising regions.

Example 17.1 - [0,1] Game #9
One full street.
One bet remaining.
The pot size is P and the bet size 1 unit.

Recall that when we solved the first finite pot [0,1] game in Chapter 11 (see Example 11.2), in addition to the yn notation we used throughout Chapter 16, we used two other notations, y_0 for the threshold between checking and bluffing, and x_1* for the threshold between X check-calling and X check-folding to a bet. In the same way that Y bluffed his very worst hands and value bet his best hands, X will likewise bluff his worst hands and value bet his best.

In the full-street game, X can additionally take the action of betting (producing a threshold x_1 and of bluffing (threshold x_0). Y has to add to his options a threshold between calling or folding if X bets, y_0*. Here, because the options are between calling and folding, it is dominated for Y to call with hands weaker than hands he would fold. Instead, he could just call with the stronger hands and fold the weaker.

We parameterize the solution this way:

X's strategy consists of the following thresholds:

- x_0 between check-folding and bluffing.
- x_1* between check-folding and check-calling
- x_1 between value betting and check-calling.

Y's strategy likewise consists of three thresholds:

- y_0 between checking and bluffing if X checks.
- y_1* between calling X's bet and folding to X's bet (if X bets)
- y_1 between value-betting and checking if X checks.

As far as ordering these thresholds, we can conjecture from our previous games that Y will bet more hands behind X's check than X would have bet - hence $y_1 > x_1$. We will also find that α, the bluffing to betting ratio (see Equation 11.1), for this game is the same as α in [0.1] Game #2 (see Example 11.2) for both players. Since the size of the bluffing region will be αx_1 for X and αy_1 for Y, we can place y_0 to the left of x_0.

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The calling thresholds for both players will be somewhere between the opponent's value betting region and his bluffing region (to make him indifferent to bluffing}. Hence we know that $x_1 < y_1^* < x_o$ and $y_1 < x_1^* < y_0$. The relationship between y_1^* and x_1^* actually turns out to be irrelevant - when we write the indifference equations for the various

thresholds, we only consider one or the other of these thresholds.

So our overall parameterization looks like this:

$0 < x_1 < y_1 < x_1^*$, $y_1^* < y_0 < x_1 < 1$

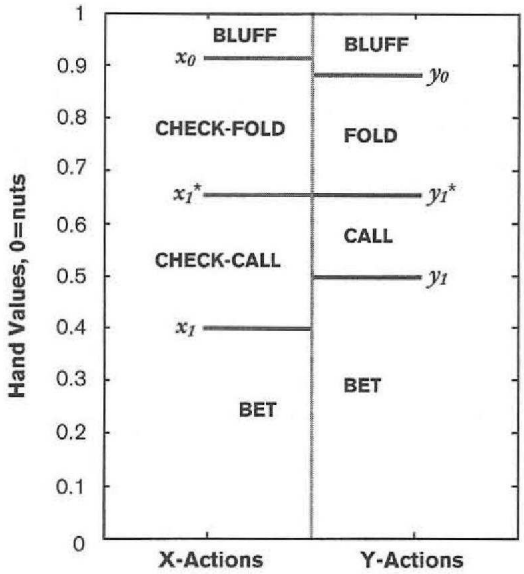

Figure 17.1, Strategy structure for [0,1] Game #9

We can now write indifference equations for each of these thresholds. We have purposely left out regions where the given action cannot occur: i.e., at y_1, Y's decision is between checking and betting. X's hands in regions $[0, x_1]$ and $[x_0, 1]$ do not allow for Y to make that decision because X has already bet; hence they are omitted.

For Y at y_1 (indifference between checking and betting):

X's hand	**Y bets**	**Y checks**	**Difference**	**Product**
$[x_1, y_1]$	-1	0	-1	$-(y_1-x_1)$
$[y_1, x_1^*]$	+1	0	+1	$x_1^*-y_1$
$[x_1^*, x_o]$	0	0	0	0
Total				$\boldsymbol{-2y_1 + x_1 + x_1^*}$

$-2y_1 + x_1 + x_1^* = 0$
$y_1 = (x_1 + x_1^*)/2$

The intuitive meaning of this equation is that Y will clearly bet all the hands that X would have bet for value. Additionally, he will bet with half the hands with which X will check call. So y_1 will be the midpoint of $[x_1\ x_1^*\}$.

For Y at y_1^*(indifference between checking and betting):

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X's hand	Y calls	Y folds	Difference	Product
$[0, x_1]$	-1	0	-1	$-x_1$
$[x_o, 1]$	+ (P+1)	0	+ (P+1)	$(P+1)(1-x_0)$
Total				$\boldsymbol{(P+1)(1-x_0) -x_1}$

$(P+1)(1-x_0) -x_1 = 0$
$1-x_0 = \alpha x_1$

This equation expresses the relationship between the size of the betting region and the size of the bluffing region. $1-x_0$ is the size of the bluffing region, and x_1 is the size of the value betting region. The multiplier α (recall that $\alpha = 1/(P + 1)$) is the ratio between the two regions.

For X at x_1* (indifference between check-calling and check-folding):

Y's hand	X check-calls	X check-folds	Difference	Product
$[0, y_1]$	-1	0	-1	$-y_1$
$[y_0, 1]$	P + 1	0	+(P+ 1)	$(P + 1)(1 - y_0)$
Total				$\boldsymbol{(P + 1)(1 - y_0) – y_1}$

$(P + 1)(1 - y_0) – y_1 = 0$
$1 - y_0 = \alpha y_1$

For Y at y_0 (indifference between check-folding and bluffing):

X's hand	Y bets	Y checks	Difference	Product
$[x_1, x_1^*]$	-1	0	-1	$-(x_1^*-x_1)$
$[x_1^*, y_0]$	+ P	0	+ P	$(P)(y_0-x_1^*)$
$[y_0, x_0]$	0	0	0	0
Total				$\boldsymbol{-(x_1^*-x_1)+(P)(y_0-x_1^*)}$

$-(x_1^*-x_1)+(P)(y_0-x_1^*) = 0$
$x_1^* - x1 = P(y_o – x_1^*)$
$x_1^* = \alpha(py_0+ x_1)$
$x_1^* = (1 - \alpha)(y_0) + ax_1 + (x_1 – x_1)$
$x_1^* = (1 - \alpha)(y_0) - (1 - \alpha)x_1 + x_1$
$x_1^* = (1 - \alpha) (y_o – x_1) + x_1$

We saw in [0,1] Game #2 (Example 11.3) that just as the bluffing to betting ratio is α, the player who has been bet into calls with 1 - α of his hands that can beat a bluff. In tills case, however, when X is deciding whether to call, he has already bet with some hands. The hands he has checked are from x_1 to x_0 - of those, only the hands better than y_0 can beat a bluff.

Then x_1* is equal to (1 - α) times the size of X's region of hands that can beat a bluff plus x_1.

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For X at x_0. (indifference between check-folding and bluffing):

Y's hand	X check-calls	X bets	Difference	Product
$[0, y_1^*]$	0	-1	+1	y_1^*
$[y_1^*, x_0]$	0	+P	-P	$(-P)(x_0 - y_1^*)$
$[y_0, 1]$	-P	0	-P	$-P(1 - x_0)$
Total				$\boldsymbol{(P + 1) y_1^* - P}$

$(P + 1) y_1^* - P = 0$
$y_1^* = P/(P+1)$
$y_1^* = 1 - \alpha$

Here we have the same situation, except that when X bets, Y has not yet limited his distribution (as X did by checking initially at the last threshold). Hence, Y calls with 1 - α of his hands. We omit the qualifier "that can beat a bluff" here, because for X the decision is between check-folding and bluffing; if he check-folds, he would lose to Y's weakest hands anyway, because Y would bluff.

For X at x_1 (indifference between check-calling and betting):

Y's hand	X bets	X check-calls	Difference	Product
$[0, x_1]$	-1	-1	0	x_1
$[x_1, y_1]$	+ 1	+1	0	0
$[y_1, y_1^*]$	+1	0	+1	$y_1^* - y_1$
$[y_1^*, y_0]$	0	0	0	0
$[y_0, 1]$	0	+1	-1	$(1- y_0)$
Total				$\boldsymbol{y_1^* - y_1 - (1- y_0)}$

$y_1^* - y_1 - (1- y_0) = 0$
$y_1^* - y_1 = 1- y_0$

Here we can see that X has two reasonable options with a hand on the cusp of initially betting or checking. When he checks, he induces bluffs and gains a bet; when he bets, he picks up a bet from Y's hands that are too weak to bet but will call a bet. Therefore, the size of these two regions is equal.

We now have six equations, each of which represents a particular relationship that must be satisfied by the optimal strategy. We also have six unknown values (the thresholds). We can solve this system of equations using algebra, to find the values of the thresholds.

$y_1 = (x_1^*+x_1)/2$
$1 - x_0 = \alpha x_1$
$1 - y_0 = \alpha y_1$
$x_1^* = (1-\alpha)(y_0-x_1) + x_1$
$y_1^* = 1 - \alpha$
$y_1^* - y_1 = 1 - y_0$

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Solution:

$y_1 = (l - \alpha)/(l + \alpha)$
$x_l = (l - \alpha)^2/(l + \alpha)$
$y_0 = (l + \alpha)^2/(l + \alpha)$
$x_0 = 1 - \alpha\ (l - \alpha)^2/(l + \alpha)$
$y_1^* = l - \alpha$
$x_1^* = 1 - \alpha$

Lets look at these thresholds in a specific example. Suppose the pot is 4 bets. Then, using Equation 11.1, $\alpha = {}^1/_5$. We know that X will value bet some fraction of his hands. Y, of course, calls with hands equal to the range with which X bluffs, breaking even on that betting, but he must also call with additional hands to make X indifferent to bluffing. When X bluffs and Y calls, he loses a single bet; when Y folds, he gains 4 bets. In order to make X indifferent to bluffing, Y must call four times as often as he folds. Since Y has the entire interval [0,1] to choose from, he calls with 4/s of his hands overall, or 1 - α.

However, X, too, must balance his value bets and bluffs so that Y cannot exploit him by calling more or less often. For example, if X never bluffed, then Y could simply call with fewer hands than X value bet, gaining value on each one. X wants to extract value with his value bets, so he must bluff in order to force Y to call with some hands between X's value betting range and X's bluffing range.

X must bluff enough so that Y is indifferent to calling with hands between x_1 and x_0. Assuming the pot is again 4, X gains 4 bets when Y folds but loses one bet when Y calls. To make Y indifferent to calling, X must value bet four times as often as he bluffs', he must bluff, then, ${}^1/_5$ as often as he value bets, or ax_1. It should also be clear by this logic that whatever hands Y value bets after X checks must be equally supported by a ratio of bluffs in the region [y_0, 1].

We then consider how often X should value bet. At the threshold point between betting and check-calling, X has what are essentially two options; he can bet, in which case he gains a bet when Y's hand is between y_2 and y_1^* (when Y calls his bet with a worse hand but would not bet if cheeked to), or he ran check, in which case he gains a bet when he induces a bluff when Y is between y_0 and 1. Since we know that X will be indifferent between these two actions, the size of these two regions [y_1 y_1^*] and [y_0, 1] must be equal.

Since we know that the second region's size is ay1 and the first region's size is $1 - \alpha - y_1$ we can easily solve for y_1 the number of hands that Y will bet if checked to. As we saw in' the solution above, this value is $(1 - \alpha)/(l + \alpha)$. Returning to the example with pot size 4, we know that y_1^* is ${}^4/_5$. We also know that $1 - y_0$ is ${}^1/_5$ of y_1. So ${}^4/_5 - y_1 = {}^1/_5 y_1$, or $y_1 = {}^2/_3$.

Now we have intuitively shown the derivations for y_1, y_1^*, y_0, and the relationship between x_1 and $1 - x_0$. Solving for x_1^* and x_1 is best done simultaneously. We find two relationships between these two quantities and then reduce them to numbers.

In the no-fold games, recall that we had a principle that when X checks, Y bets half the hands that X would check (provided that X cannot raise). In the finite pot game, however, when Y is deciding to check or bet at y_1 he gains a bet when X is between y_1 and x_1^* and loses a bet when X is between x_1 and y_1. These two regions must be of equal size, so y_1 must be the midpoint between x_1 and x_1^*.

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We can restate this as:

If your opponent could have bet but checks, you cannot be raised, and your opponent may call or fold, the proper strategy is to value bet all hands that your opponent would have value bet and half the hands with which he will call.

We can see that this dovetails with the earlier principle - in the no-fold case, the opponent must always call. Hence, value betting half the hands is correct. However, if your opponent may fold, the idea is to bet half the hands he will call.

Additionally, X must choose x_1* such that he defends against Y's bluffs in the same way that Y defended against X's bluffs. However, X does not have the entire [0,1] distribution for defense; instead, he has only the subset [x_1, x_0], as he would have bet hands outside of this range, either as a bluff or as a value bet. Still, though, he must defend against Y's bluffs. So he must call with (1 - α) of the region [x_1, x_0]. This implies that x_1* must be 1 - α of the way from x_1 to x_0.

With pot size 4, $y_2 = {}^2/_3$ is the midpoint of x_1 and x_1* while x_1* is ${}^4/_5$ of the way from x_1 to 1 - (${}^1/_5$)x_1. If we call *c* the distance from x_1 to x_0, a the distance from x_1 to y1 and b the distance from 0 to x_1, we get the following relationships:

$b = {}^2/_3 - \alpha$
$1 = {}^4/_5 b + {}^5/_2 \alpha$
$b = {}^4/_5 (1 - {}^5/_2 \alpha) = {}^4/_5 - 2\alpha$
${}^4/_5 - 2\alpha = {}^2/_3 - \alpha$
$\alpha = {}^2/_{15}$
$b = {}^8/_{15}$
$c = {}^1/_3$

.. .which matches the results we get from plugging into the formulas (b = x_1).

Understanding this game is essential for moving on to more complex finite pot games. We spent a fair amount of time explaining intuitively the meanings of the equations. As we move forward, many things that were explained here will be taken for granted, so we encourage you to take some time to try to work back through how the intuitive explanations mirror the algebra above. The next game is the counterpart to [0,1] Game #6 (Example 16.3) - the introduction to raising.

Example 17.2 - [0,1] Game #10

One full street.
Two bets left.
Check-raising is not allowed.
The pot size is P and the bet is 1 unit.

You may have already guessed that in this game we will again utilize the y_2 notation. Additionally, we will introduce the last piece of notation necessary to generalize the [0,1] game - the raise-bluff notation.

Consider the situation of Player Y after X has bet. What should his strategy look like? It should be intuitively clear by now that in α certain region (his best hands), he will raise for value; with a different, weaker region of hands, he will call to force his opponent to bluff correctly, and with yet another region he will fold because his hands lack the necessary value to continue.

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Suppose we set up a strategy with these three regions. Now consider the situation after value-raises. Given a strategy of the type above, how should X respond? X knows that Y has a strong hand for a raise, so he can fold hands beneath Y's raising threshold and equalize. This is analogous to what X would do if there were only one bet left and Y never bluffed; X would simply throw away his mediocre hands. So Y cannot just value raise with his best hands without protecting those value raises with some type of raise-bluffs.

The next question is: From where should those bluffs come?

To answer this question, we must consider the nature of a bluff. In cases where there is only one bet remaining, we bluff with our very worst hands. We choose the very worst hands because they are the hands that have the least showdown value if the pot is checked through. These hands stand to gain the most from being played as a bluff. When the opponent calls our bluffs, we should never win the pot if the opponent is playing optimally (or even just rationally).

But now consider the situation when the opponent has already bet. We will still want to raise with our best hands and call with our next group of hands because we want to use the showdown value of our hands most efficiently. Using the principle that raise-bluffs should never win the pot if called, we can see that the raise-bluffs will come from the remaining fraction of hands that are neither raising hands nor calling hands. Of that fraction, some will be raise-bluffs, m a proportion that protects our value raises properly with respect to the size of the pot and the size of our value-raising region, and the remainder will be folded.

Perhaps the reader's intuition now leads to a situation similar to bluffing in one-bet scenarios, where the bluffs are the worst hands in the region. But this is incorrect. Technically speaking, for Y the raise-bluffs can come from anywhere in the group of hands worse than y_1* and be at least co-optimal. However, since we are going to fold the non-raise-bluff hands and there is no hope of a showdown with hands better than a bluff, using the strongest hands in the (fold, raise-bluff) region dominates any other strategy.

This gives rise to a new type of threshold and also forces us to redefine a previous threshold for games with more than one bet.

- y_n* is redefined as the threshold between calling the nth bet and, if the (n + l)th bet and folding are allowed, putting that bet in as a bluff, or otherwise folding.
- y_n# is the threshold between the region which puts the nth bet in as a bluff and the region which folds to the $(n - l)^{th}$ bet.

As a result of these definitions, y_n* will be paired and closely related to $_{yn+1}$#.

When we perform parameterizations, different game trees are sometimes possible based on previous actions. As a result, sometimes we may draw thresholds above or below other ones when the other is not necessary, because the two are never material at the same time.

For example, x_1* only matters if X checks. If X does check, then y_1* and y_2# are irrelevant. So whether x_1* lies above or below y_1* is arbitrary for the parameterization (although the solution will still tell the location).

With the above caveats, we parameterize [0,1] Game #6 this way:

0 ,y_2, x_2*, x_1, y_1, x_1*, y_1*, y_2#, y_0, x_0, 1

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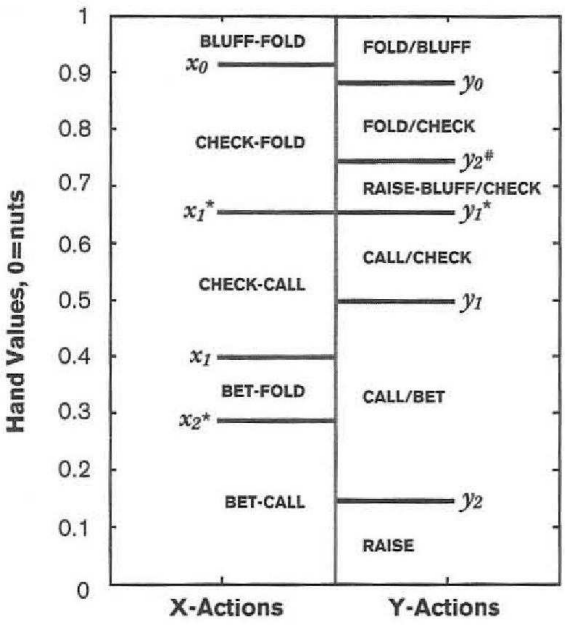

Figure 17.2. Strategy structure for [0,1] Game #10

Translating this parameterization into strategies:

For X:

X bets for value and calls a raise on [0, x_2*).
X bets for value and folds to a raise on [x_2*, x_1],
X checks and calls a bet on [x_1, x_1*].
X checks and folds to a bet on [x_1*, x_0].
X bluffs on [x_0 1],

For Y:

If X bets:

Y raises for value on [0, y_2].
Y calls on [y_2, y_1*]
Y raise-bluffs on [y_1*, y_2#]
Y folds on [y_2#, 1],

If X checks:

Y bets for value on [0, y_1].
Y checks on [y_1, y_0]
Y bluffs on [y_0, 1].

Again, we advise you to carefully review this notation. Understanding what the threshold values mean is of great significance to understanding this and more complex games. In later games, we will see values such as, x_4#, which is the threshold between X check-raising and putting in the fourth bet as a bluff and X folding to the third bet after check-raising, and it is easy to become lost as a result of not mastering the notation.

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In the one-bet case, we introduced a variable a to represent the fundamental ratio$1/(P + 1)$ that governs in one way or another the relationships between betting and bluffing, as well as calling. In two-bet cases, however, we have situations where there is a larger pot already, and the bluff itself costs two bets. Hence, we will have a different ratio.

$\alpha_2 = 1/(P + 3)$

In other games, we will use this notation for larger number of bets than two, so more generally:

$\alpha_n = 1/(P + (2_n - 1))$ **(17.1)**

The α from [0.1] Game #2 (Example 11.3), then, is more strictly α_l. However, we will use the convention that an unmarked α is α_1.

To solve this game, we now have nine unknowns. At each of these unknowns we will have an indifference equation, and solving the system of indifference equations will yield our solution.

For Y at y_2:

X's hand	Y calls	Y raises
$[0, y_2]$	-1	-2
$[y_2, x_2^*]$	+1	+2
$[x_2^*, x_1]$	+1	+1
$[x_0, 1]$	+1	+1

Y loses an extra bet by raising X's best hands, and gains a bet from raising hands worse than y_2 that will call. Since X cannot raise, we see the familiar idea of Y raising with half the hands that X will call.

$y_2 = x_2^* - y2$
$y_2 = x_2^*/2$

For Y at y_1, we have exactly the same situation as in [0,1] Game #9 (Example 17.1), since X cannot raise. Hence, our indifference equation here is the same.

$y_1 = (x_1^* + x_1)/2$

y_1 is still the midpoint of X's betting thresholds and X's calling threshold.

For Y at y_1^*:

X's hand	**Y calls**	**Y raise-bluffs**
$[0, x_2^*]$	-1	-2
$[x_2^*, x_1]$	-1	+(P+1)
$[x_0, 1]$	+1	+1

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$x_2^* = (P + 2)(x_1 - x_2^*)$
$x_2^* = x_1(P + 2)/(P+ 3)$
$x_2^* = (1 - \alpha_2)x_l$

This is the first appearance of the idea of a_2. Here we see that the size of X's betting region x_1 is related to the size of the region where he bets and calls a raise. In fact, he bets and folds to a raise α_2 of the time. This is because he must make his opponent indifferent to raising as a bluff. α_2 is used instead of a because the pot is now larger - X's bet and Y's raise are both now in the pot.

For Y at y_2#:

X's hand	Y raise-bluffs	Y folds
$[0, x_2^*]$	-2	0
$[x_2^*, x_1]$	+(P + 1)	0
$[x_0, 1]$	+1	- P

First, a bit of algebra:

$2x_2^* = (P + 1)(x_1 – x_2) + (P + 1)(1 – x0)$
$2x_2^* = (P + 1)(x_1 - x_2^* + 1 - x_0)$
$(P + 3)\, x_2^* = (P + l)(x_l + l - x_0)$
$\alpha x_2 = \alpha_2(x_1 + l - x_0)$
$\alpha(l - \alpha_2)x_1 = \alpha_2(x_1 + 1 - x_0)$
$(\alpha/\alpha_2 - \alpha)x_1 = x_1 + 1 - x_o$

We can also create an identity:

$\alpha/\alpha 2 = (l/(P+ l))/(l/(P+3))$
$\alpha/\alpha 2 = (P+3)/(P+l)$
$\alpha/\alpha 2 = 1 + 2\alpha$

Substituting for $\alpha/\alpha 2$:

$(1 + \alpha)x_1 = x_1 + 1 - x_0$
$\alpha x_1 = 1 – x_0$

Remember that previously we commented that Y's raise-bluffing region could really be anywhere in his folding region. We insist on playing non-dominated strategies, so as a result, we get an indifference between raise-bluffing and folding. However, we could also solve the game for a different threshold between calling and folding; then the above equation would simply appear instead of forcing us to go through algebraic manipulations.

This is the familiar equation (e.g., Equation 11.4, among others) that relates the size of X's betting region to the size of his bluffing region by a factor of α.

For Y at y_0, we have the same equation as in [0,1] Game #9 (Example 17.1), because X still cannot check-raise.

$x_1^* = (1 - \alpha)(y_0 – x_1) + x_l$

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For X at x_2*:

Y's hand	X bet-calls	X bet-folds
$[0, y_2]$	-1	-2
$[y_1^*, y_2\#]$	+2	-(P+1)

Notice that X's strategies are more complicated in this game. When we examine the indifference between two actions, it is not enough to state "X bets" or "X checks" - instead we must specify the entire strategy :'X bet-calls" or "X bet-folds."

x_2* is one of the new thresholds, between X calling a raise and folding to the raise. X loses a bet by calling against Y's value raises, but picks off Y's raise bluffs. As we shall see, because it costs Y two bets instead of one to make a raise-bluff, the calling ratio for X is differently shaped.

$y_2 = (P + 3)(y_2\# - y_1^*)$
$\alpha_2 y_2 = (y_2\# - y_1^*)$

This equation is basically the raise-bluff equation analogous to Equation 11.4, but for two bets instead of one from earlier games. The size of the raise-bluffing region $y_2^* - y_1\#$ is the size of the value raising region (y_2) multiplied by the appropriate ratio (α_2). This is a general rule that appears throughout all the [0,1] finite pot games. For every value region, there is a corresponding bluffing region. The size of the bluffing region is related to the size of the value region by some αn value.

For X at x_1:

Y's hand	X bet-folds	X check-calls
$[0, x_1]$	-1	-1
$[x_1, y_1]$	+1	+1
$[y_1, y_1^*]$	+1	0
$[y_1^*, y_2\#]$	-(P + 1)	0
$[y_2\#, y_0]$	0	0
$[y_0, 1]$	0	+1

$y_1^* - y_1 = (P+ l)(y_2\# - y_1^*) + (1 – y_o)$

At x_1, we have an equation that relates the size of the region where Y calls a bet but wouldn't have bet himself ($y_1^* - y_1$), and the size of the raise-bluffing and bluffing regions for Y. This is X's choice between trying to induce a bluff with his borderline value betting hand or trying to extract value. If he bets, he gains a bet when Y calls but wouldn't have bet, but loses P+ 1 bets when Y raise-bluffs him out of the pot. If he checks, he sometimes induces a bluff from Y. The value of checking and the value of betting must be equal; this equation reflects that.

For X at x_1*, we again have the familiar relationship:

$1 – y_o = \alpha y_1$

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For X at x_0:

Y's hand	X check-folds	X bluffs
$[0, y_2\#]$	0	-1
$[y_2\#, x_0]$	0	+P
$[x_0, 1]$	-P	0

$y_2\# = (x_0 - y_2\#)(P) + (1-x_0)(P)$
$(P + 1)y_2\# = P$
$y_2\# = 1 - \alpha$

This result is somewhat analogous to the equation governing $y_1\#$ in [0,1] Game #9 (Example 17.1); the total hands that Y folds to X's bet with is α. In [0,1] Game#9 we characterized this relationship as "Y calls with (1 - α) of his hands," but a more accurate characterization in light of this result is "Y folds a of his hands to a bet."

We now have nine equations. Many of them are the same as the previous game.

$y_1 = (x_1^* + x_1)/2$ **(17.2)**

$y_2 = x_2^*/2$ **(17.3)**

$x_2^* = (1 - \alpha_2)x_l$ **(17.4)**

$\alpha x_1 = 1 - x_0$ **(17.5)**

$x_1^* = (l - \alpha)(y_0 - x_1) + x_1$ **(17.6)**

$\alpha_2 y_2 = (y_2\# - y_1^*)$ **(17.7)**

$y_1^* - y1 = (P+ 1)(y_2\# - y_1^*) + (1 - y_0)$ **(17.8)**

$l - y_0 = \alpha y1$ **(17.9)**

$y_2\# = l - \alpha$ **(17.10)**

We can solve this system of equations and obtain values for all the thresholds. We will (hopefully to your pleasure) omit much of the algebra, but there is at least one point worth a bit of additional scrutiny.

Combining (17.3) and (17.4), we have the important relationship:

$y_2 = x_1(1 - \alpha_2)/2$ **(17.11)**

As in [0,1] Game #6 (Example 16.3), we can rewrite this to capture its identity as the ratio between the hands that X bets and the hands that Y raises:

$y_2 = Rx_1$

Here, $R = (1 - \alpha_2)/2$. We can see that as P goes to infinity, this value converges on $^1/_2$, which was the value that it took in [0,1] Game #6.

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The idea behind solving this system of equations is to convert everything to be in terms of y_1 and x_1. We can solve Equations 17.2 and 17.6 for x_1*, set them equal, and obtain one equation that contains only x_1 and y_1. A second equation can be obtained by working on Equation 17.8. This equation contains y_1*,y_1, y_2#, and y_0.

y_0 can be put in terms of y_1 with 17.9. y_2# is a constant by 17.10. We can put y_1* in terms of y_1 by using 17.7. can be put in terms of x_1, as $y_2 = Rx_1$. Since y_2# is a constant, y_1* is now in terms of x_1. Substituting for all these variables can make 17.8 a second equation in terms of only x_1 and y_1.

We can reduce these two equations to the following:

$$x_1 = (l - \alpha)^2/[l + \alpha + (2 - \alpha)(l - a_2)R] \qquad \textbf{(17.12)}$$

This equation is very important for solving games without check-raise, because it is true for not only the two-bet game but games with more bets where folding is allowed but check-raising is not. This is because the addition of more bets changes the value of R, but does not change the fundamental relationships among the lower thresholds. We will see that this relationship greatly simplifies the infinite bets case addressed below.

Once we have this, we can solve for y_1 by substitution and the remaining variables are easily found using Equations 17.2 through 17.10.

Solution:

$y_2 = (l - a)^2(l + 2a) / (7a^2 + 9a + 4)$
$y_1 = (1 - a)(8a^2 + 9a +3)/(l + a)(7a^2 + 9a + 4)$
$y_1* = 1 - a - a_2((l - a)^2(l + 2a) / (7a^2 + 9a + 4))$
$y_2\# = 1 - a$
$y_0 = 1 –a(1 - a)(8a^2 +9a+3)/(l + a)(7a^2+9a + 4)$
$x_2* = 2(1 – a)^2(1+ 2a) / (7a^2 + 9a + 4)$
$x_1 = 2(1 - a)^2(l + 2a)2/(1 + a)(7a^2 + 9a + 4)$
$x_1*= 4(1 - a)(2a^2 + 2a + 1)/ (7a^2 + 9a + 4)$
$x_0 = 1 -2a(l -a)^2(l + 2a)^2/ (1 + a)(7a^2 + 9a + 4)$

This is the solution to what is effectively the simplest complete two-bet [0,1] game. Admittedly, this final solution isn't particularly useful - what is useful is the method and the principles that were used in deriving it.

We can look at what the thresholds look like for this game for different pot sizes. The rightmost column in the following table contains a pot size of ε, a value frequently used by mathematicians to represent a tiny quantity. Here we use it to show that as the pot converges on zero, the thresholds converge to "no betting" as we expect.

Threshold	**P = ∞**	**P = 1**	**P = 2**	**P = 9**	**P = ε (~0)**
y_2	$^1/_4$	$^2/_{41}$	$^2/_{21}$	0.195573441	0
x_2*	$^1/_2$	$^4/_{41}$	$^4/_{21}$	0.391146881	0
x_l	$^1/_2$	$^{16}/_{123}$	$^5/_{21}$	0.426705689	0
y_1	$^3/_4$	$^{76}/_{246}$	$^{279}/_{630}$	0.655203951	0
x_1*	1	$^{20}/_{41}$	$^{68}/_{105}$	0.883702213	0

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y_1*	1	$^{20}/_{41}$	$^{68}/_{105}$	0.883702213	0
y_2#	1	$^{1}/_{2}$	$^{2}/_{3}$	0.9	0
y_0	1	$^{208}/_{246}$	$^{537}/_{630}$	0.934479605	0
x_0	1	$^{230}/_{246}$	$^{58}/_{63}$	0.957329431	0

We continue our discussion with the two-bet game where X can check-raise:

Example 17.3 - [0,1] Game #11

One full street.
Two bets left.
Check-raising is allowed.
The pot size is P and the bet is 1 unit.

As we progress through additional games, we will generally omit listing off the whole solution to the game and calculating each threshold value. As you saw in [0,1] Game #10 (Example 17.2), the solutions to more complex games with finite pots are substantially more complex and not that useful for seeing at a glance. Instead, we will focus on building on the results we have from previous games and showing the areas in which the strategies for more complex games differ from differ less complex counterparts.

In the check-raise game with two bets remaining, we have a number of additional thresholds. However, many of these are easily dismissed because they relate to each other m familiar ways. For example, X will have a check-raise bluff region between x_2# and x_1*. We know from our experience solving indifference equations that the size of this region will be the size of the value check-raising region (x_2) multiplied by the appropriate α value α_2- Likewise, instead of just a value betting region to the left of y_1, Y will have a "bet-call" region to the left of y_2* and a "bet-fold" region between y_2* and y_1. These will be sized such that Y folds 1 - α_2 of his value bets to a raise. This makes X indifferent to raise-bluffing, and so on.

"What's interesting and important here, though, is the change in the value regions for both players. Changes in these value regions will cascade through the other thresholds and impact both players' strategies, but the relationships of the other thresholds to the value regions are fairly simple and straightforward. We could apply the method of writing indifference equations for each threshold, but instead let us simply draw attention to the important thresholds.

In the check-raise game, we know that Y's value-betting strategy after X checks will be impacted by X's ability to check-raise; additionally Y's value-raising strategy will be impacted by the fact that X is checking some of his strong hands and likely betting a little weaker than he does in the game with no check-raise. Recall that these were the major shifts in the game with no folding allowed - Y bet a little tighter after X checked (fearing a check-raise), X bet a little looser (to capture some of the value lost by not inducing weaker hands to bet), and Y raised a little looser (because of X's looser betting standards).

There is an important link between the check-raise game and the no-check raise game. To see this, consider the indifference at x_2.

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For X at x_2:

Y's hand	X check-raises	X bets-calls
$[0, x_2]$	-2	-2
$[x_2, y_2]$	+2	+2
$[y_2, y_2^*]$	+2	+1
$[y_2^*, y_1]$	+1	+1
$[y_1, y_1^*]$	0	+1
$[y_1^*, y_2\#]$	0	+2
$[y_0, 1]$	+1	0

$y_2^* - y_2 + (1 - y_0) = 2\,(y_2\# - y_1^*) + y_1^* - y_1$

To find y_2^*, we consider the indifference at $x_2\#$:

Y's hand	X check-calls	X check-raise-bluffs
$[0, y_2^*]$	-1	-2
$[y_2^*, y_1]$	-1	-(P+ 1)
$[y_0, 1]$	+ 1	+i

$y_2^* = (P + 2)\,(y_1 - y_2^*)$
$y_2^* = (1 - \alpha_2)y_1$

We additionally know that $y_2\# - y_1^*$ is equal to $\alpha_2 y_2$.

So we have:

$(1 - \alpha_2)y_1 - y_2 + (1 - y_0) = 2\alpha_2 y_2 + y_1^* - y_1$

In the no-check-raise game, we had the following from the indifference at x_1

$y_1^* - y_1 = (P + 1) + (y_2\# - y_1^*) + (1 - y_o)$

The origin of this equation is that X must be indifferent to bet-folding and check-calling at x_1. When he bets, he gains a bet against Y when Y has a hand that will call a bet but not bet if checked to. Against this, he loses (P + 1) bets when Y raise-bluffs, and additionally loses the bluffs that he fails to induce by betting. The bluff terms drop off, and we have:

$(1 - \alpha_2)y_1 - y_2 = (P + 3)\alpha_2 y_2$
$(1 - \alpha_2)y_1 = 2y_2$
$y_2 = y_1(1 - \alpha_2)/2$

Recall that in the no-check-raise game, we saw that this ratio $(1 - \alpha 2)/2$ was the ratio between y_2 and x_1. Here we see that it is the ratio between y_2 and y_1. Likewise, when we solved the infinite bets game without check-raise (Example 16.3), we saw that r was the ratio between successive thresholds (y_2, x_3, y_4, etc.), while in the game where check-raise was allowed (Example 16.4), r is the ratio between successive y_n thresholds.

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As a general result, we find that when converting [0,1] games from no check-raise to including check-raise, this relationship holds (for simple symmetrical cases). The ratio of Y thresholds to X thresholds in the no-check-raise game is the ratio of successive Y thresholds in the game with check-raise.

We can follow this chain of logic to find that the general equation for y_1 in this game is:

$$y_l = (1 - \alpha)/[l + \alpha + (1 - \alpha_2)R] \qquad \textbf{(17.13)}$$

This equation is directly analogous to Equation 17.12, but includes the check-raise element of this game.

Calculating the values of the various thresholds is a straightforward process of creating indifference equations and solving them. Bluffing regions have size related by an a value to their corresponding value regions, and so on.

hi the next game, we tackle the more complex problem of allowing infinite raises.

Example 17.4 - [0,1] Game #12

One full street.
Check-raise is not allowed.
An arbitrary number of bets and raises are allowed.
Folding is allowed.

In solving this game, we will relate its components to games we have previously solved. Most of the indifference equations in this game are identical to then counterparts in [0,1] Game #10 (Example 17.2); in fact, the only indifference equation that is invalid is the indifference at y_2. In that game, the second bet was the last one; hence, Y could not be raised back. We found there that Y raised with half the hands with which X would call. However, here X can additionally raise back, so we must reconsider that equation.

Additionally, we add an infinite number of thresholds to the game, allowing for the arbitrary allowed bets and raises. For each threshold, we have an additional indifference equation. Many of these equations have a clear form, as for example we will obtain for any appropriate *n*:

$$a_n y_n = (y_n\# - y_{n-1}*)$$

This is the familiar bluffing-to-betting ratio (based on Equation 11.4), governed by the appropriate α. value and the number of bets to be put in the pot.

When we solved [0.1] Game #10, we identified a key equation that is the general form of x_1 for non-check-raise games.

$$(1 - \alpha)^2 = x_1[l + \alpha + R(2 - \alpha)(l - \alpha 2)]$$

The equation that must be modified to allow for the difference between this game and [0,1] Game #10 is the following:

$$y_2 = Rx_1$$

However, this equation need not change except that *R* has a new value.

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Hence, Equation 17.12 still holds. All that's necessary is that we find the appropriate *R*. We can check this by looking at games we have already solved.

Consider the game where folding was not allowed (Example 16.4). Allowing the pot to go to infinity, we can see that all the α values go to zero. In that game, y_2 was $^1/_2$ and x_1 was $^1/_2$. Hence, R was $^1/_2$. Plugging this into our formula and letting α and α2 go to zero, we get $x_1 = {}^1/_2$.

In the one-bet game, R was zero because y_2 was at 0. So x_1 equals $(1 - \alpha)^2/(l + \alpha)$. Looking back to the solution for [0,1] Game #9, we see that this is the case.

Returning to the game at hand, we find the following indifferences (which should be intuitive to those who have come this far):

$x_3\# - x_2^* = \alpha_3 x_3$

The size of the nth bluffing region is αn times the size of the corresponding value region.

$x_3\# = (1 - \alpha_2)x_1$,

When you put in the nth bet for value, you must at least call (or raise) with $1 - \alpha_{n+1}$ of the hands that originally put in that bet (to avoid being exploited by bluff raises).

We also have the following indifference at y_2. (This is the lowest threshold at which this game differs from [0,1] Game #6.)

$x_2^* - y_2 = y_2 + (P + 3)\alpha_3 x_3$

Y picks up a bet at y_2 by value raising when X is between y_2 and x_2^*. He loses one additional bet whenever X is better than y2 (folding to additional raises) and also loses (P+ 3) bets when X is in his reraise-bluff region $x_3\# - x_2^*$.

Solving this equation, wy find:

$(1 - \alpha_2)x_1 - \alpha_3 x_3 = 2y_2 + (P + 3)\alpha_3 x_3$

Converting α terms to include P:

$x_1(P + 2)/(P+ 3) = 2y_2 + x_3(P + 4)/(P+ 5)$ **(17.14)**

The indifferences here are identical as we increment the thresholds by two bets; hence this equation is a particular case of the general recursion for x_n, y_{n+1}, x_{n+2} as we increment to higher thresholds.

We can take advantage of a principle that once we reach the second raise, the game becomes completely symmetrical in terms of the ratios between thresholds except for the size of the pot. The values of the thresholds are different, of course, because they have been multiplied by the value analogous to *r*, the same value as in Equation 14.1. We have been calling this value R; because to this point, we have only considered games where one raise was allowed. However, as the pot grows, the value of this multiplier changes as well.

We will then call this changing multiplier R_p because it changes based on the size of the pot, and we have the following:

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$y_2 = R_p x_1$
$x_3 = R_p + 2y_2$
$y_4 = R_p + 4x_3$

...and so on.

Substituting into 17.14:

$x_1(P+2)/(P+3) = 2R_p x_1 + (R_p R_p + 2x_l)(P+4)/(P+5)$

The x_1 terms drop out, and we have:

$$R_p = \frac{\frac{(P+2)}{(P+3)}}{2 + \frac{(R_{p+2})(P+4)}{(P+5)}}$$

This is the general recursion that describes the relationship between successive values of R_p. To find specific values, we can use the fact that we know that R_p converges on r as P goes to infinity (since this is an approximation of the no-fold game as shown in Example 16.4). Hence, we can choose an arbitrarily large value of n, set it equal to $\sqrt{2}$ - 1, and work backwards to the desired P. As it happens, this recursive relation converges quite rapidly. Of course, we can see that for the no-fold case, R_p simply is r for all thresholds, and this game simplifies to its no-fold analogue.

Once we have the value of R_p, we can easily calculate the values of the thresholds by solving for x_1 and then multiplying by the appropriate R_p values. As usual, the bluffing regions and calling regions follow the value regions, and we have a solution.

Key Concepts

- The [0,1] games can be extended to cover more complicated situations, including check-raising, an arbitrary number of bets, and so on.
- In all of these situations, however, the keys to the game are the value thresholds. For each value region, there are corresponding bluffing, calling, folding, and raise-bluffing regions that can be calculated by relating the size of the pot to the size of the value region.
- In games without check-raise, there is a recurrence relation between alternating X and Y thresholds such that $x_1 -> y_2 -> x_3$ and so on.
- In games with check-raise, there is a similar recurrence relation between consecutive Y thresholds.
- If your opponent could have bet but checks, you cannot be raised, and your opponent may call or fold, the proper strategy is to value bet all hands that your opponent would have value bet and half the hands with which he will call.

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